Sensors in a Formula Student Racecar

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Abstract

Sensors help a Formula Student racecar measure its movement, speed, position and surroundings. This report explains how the main sensors work, where they can be useful to Formula Manipal, what they may cost, and where they can be sourced online in India. It covers the inertial measurement unit (IMU), satellite positioning (GNSS), wheel speed sensors, mono and stereo cameras, LiDAR, and supporting sensors used during vehicle testing. Prices are indicative examples from online listings and can change with stock, tax and shipping.

1. Introduction

A racecar needs reliable measurements to understand how it behaves on track. Wheel speed helps assess traction, an IMU records acceleration and turning, and suspension sensors show how the car moves over bumps. These measurements let the team compare setup changes instead of relying only on driver feedback. For example, if the car feels unstable while braking, wheel speed and IMU data can help show whether the wheels slow unevenly or the car pitches forward sharply. Data collected over several laps can also reveal patterns that are difficult to notice during a single run.

The right choice depends on the team's budget, whether the car is driver-operated or driverless, and how much detail is needed for testing. Low-cost sensors can support basic data logging, while higher-grade sensors usually provide better accuracy, stronger construction and more dependable readings. For a driverless car, cameras and LiDAR can help identify cones and track boundaries. For a driver-operated car, sensors used for performance analysis may offer more immediate value. Each sensor should be selected for a clear purpose, rather than added simply because it is available.

2. Main Vehicle Sensors

A. Motion Sensor (IMU)

An inertial measurement unit (IMU) measures acceleration and how quickly the car turns. Most units combine accelerometers, which measure changes in speed along three directions, with gyroscopes, which measure turning rate. Some also include a magnetometer that measures the Earth's magnetic field. Together, these readings help estimate the car's movement and orientation. This helps the team study cornering, braking, vibration, vehicle balance and how quickly the car responds to steering. IMU data can also be compared with wheel speed and GPS data to check whether different sensors tell a consistent story.

The sensor should be fixed firmly to the car, preferably near its centre, and positioned in line with the car's direction. A loose mount can add vibration that looks like real vehicle movement. The team should record the sensor's orientation and check its readings while the car is stationary before testing on track. Low-cost boards such as the MPU-6050 and BNO055 are easy to find through Indian electronics stores, with example listings ranging from roughly ₹140 to ₹1,500 depending on the board and seller. These are useful for learning and early testing, but their accuracy and vibration performance should be checked before they are trusted for detailed vehicle analysis.

B. Satellite Positioning (GNSS and RTK)

A GNSS receiver uses signals from satellites to estimate the car's position and speed. GPS is one part of GNSS; many receivers can use more than one satellite system, which can improve the number of available signals. A basic module is useful for logging the route, lap timing and approximate speed. However, ordinary GNSS can be off by several metres, so it may not reliably distinguish between nearby cones or accurately show the car's line through a corner. The antenna needs a clear view of the sky and should be mounted away from electrical noise where possible.

Real-Time Kinematic (RTK) positioning improves accuracy by using correction data from a base station or correction service. With a suitable receiver, antenna and correction link, RTK can reach centimetre-level accuracy in good conditions. This can help map laps, compare racing lines and support driverless navigation. It is more expensive and requires more setup than a basic GPS module; trees, buildings, poor antenna placement and weak correction coverage can reduce performance. Entry-level GNSS modules listed by Indian retailers may cost around ₹400–₹1,300, while higher-performance modules can cost several thousand rupees or more. RTK-capable kits are typically much more expensive, so the team should confirm the exact model, antenna and correction requirements before buying.

C. Wheel Speed Sensors

Wheel speed sensors measure how fast each wheel rotates. A common type detects a toothed or magnetic ring as it turns and sends pulses to the data system. The controller counts these pulses over time and converts them into wheel speed. Comparing the four wheels can help the team identify wheelspin, wheel lock during braking, or differences caused by tyre size and pressure. This information is useful when evaluating traction, braking behaviour and the effect of setup changes. Wheel speed can also help estimate vehicle speed, although tyre slip means it should not be treated as a perfect measurement of the car's movement.

Hall-effect sensors are a practical option because they can measure rotation without physical contact and can work at low speeds when correctly selected. Optical sensors are another

option, but they may be affected by dirt, water or blocked light. The sensor needs a secure bracket, a suitable target ring and the correct gap between the sensor and the target. The team must also protect wires from heat, moving parts and vibration. Basic hobby encoder modules can cost under ₹200, but a racecar-ready sensor and connector may cost much more. Before purchase, confirm the output type, operating voltage, maximum speed, mounting dimensions and whether the signal can be read by the team's data logger.

3. Cameras and LiDAR

A. Cameras

Cameras provide images that software can use to identify cones, track edges and other objects. A mono camera uses one lens and produces a normal two-dimensional image. It is usually cheaper, lighter and simpler to install, and software can use colour, shape and position in the image to detect cones. However, a mono camera does not directly measure distance from one image. Distance must be estimated using known object size, camera calibration or movement between frames, and these methods can be affected by viewing angle and lighting.

A stereo camera uses two cameras placed a fixed distance apart. By comparing where an object appears in each image, software estimates depth in a way similar to human vision. Stereo cameras can provide useful distance information for cone detection and path planning, but they need careful calibration and enough visual detail in the scene. Plain surfaces, glare, shadows and low light can make depth estimates less reliable. A wide view can capture cones beside the car, while a narrower view can help detect objects farther away. The Stereolabs ZED 2i is an example of a stereo depth camera; its price is much higher than a basic USB camera, so the team should compare field of view, frame rate, resolution, compute needs and environmental protection before choosing one.

B. LiDAR

LiDAR sends out laser light and measures how long the reflected light takes to return, allowing the sensor to estimate distance. A scanning unit measures many points around it and can build a view of nearby objects. This is useful for driverless software because cones and track boundaries can be located even when their colours are difficult to separate from the background. Unlike a camera, LiDAR provides distance measurements directly, but it does not usually provide the same colour and visual detail as an image. Some low-cost modules measure distance in only one direction, while scanning units cover a wider area.

LiDAR is usually mounted near the front of the car with a clear view of the track. It must be protected from vibration, dirt and possible impacts, and its position should be measured accurately so software can relate the scan to the car. Small distance modules such as the Benewake TF-Luna or TFMini-S are listed by Indian sellers for roughly ₹1,800–₹4,000, depending on model and seller. These single-point sensors are useful for short-range experiments but are not a substitute for a full scanning unit. Two-dimensional scanning LiDAR products can cost around ₹8,000–₹53,000 in online Indian listings, while automotive-grade units can cost far more. The team should check range, scan rate, interface, field of view and outdoor performance before selecting a model..

C. Choosing Between Mono, Stereo and LiDAR

The three perception options solve related but different problems. A mono camera is usually the least expensive way to start because it captures visual information using one camera. It is useful when the software mainly needs to recognise cones and estimate their location in the image. Stereo vision adds depth information by comparing two views, but it requires more processing and good calibration. LiDAR provides distance points directly and can work well when colour information is weak, but scanning units are generally more expensive and may produce less descriptive information about an object. For a student team, the decision should be based on the required driving speed, cone spacing, lighting conditions, computer capacity and budget.

A practical comparison should use the same test track and objects for every option. The team can place cones at known distances, record sensor output, and check how often the software detects them correctly. Repeat the test in bright sunlight, shade and different cone positions. This gives a clearer basis for selecting equipment than relying only on advertised range or resolution. It is also important to check the full system: a high-quality camera may not perform well if the computer cannot process its images quickly enough.

4. Supporting Sensors for Vehicle Testing

Other sensors help measure parts of the car that the main sensors do not cover. Linear position sensors can measure suspension movement, ride height, steering movement and pedal travel. A steering angle sensor records how far the steering wheel or steering rack has turned, helping the team compare driver input with the car's response. Brake pressure sensors can show how pressure builds during braking and help identify differences between runs. These sensors are most useful when the team wants to understand why the car behaves in a particular way, rather than only measuring the final speed or lap time.

Temperature and pressure sensors can monitor engine, motor, battery, tyre or cooling-system conditions, depending on the car's powertrain. For an electric racecar, battery temperature and voltage monitoring are important for checking that the system stays within its permitted operating limits. For a combustion car, oil pressure, coolant temperature and other engine measurements can help detect problems early. These sensors must be selected for the correct range and fluid or electrical environment. Pedal-position sensors should be installed and checked carefully because their readings may be connected to control or safety systems. Sensor data used for safety decisions should be validated and should not replace required mechanical checks.

A. *Data Logging and Sensor Calibration*

Sensor readings become more useful when they are recorded together with accurate timestamps. For example, wheel speed, IMU acceleration and GNSS position should refer to the same moment in time. If one sensor is delayed, the team may wrongly connect a wheel-speed change to a cornering event. The logger should record units, sampling rate and sensor status, and it should make missing readings easy to identify. A short pre-run check can confirm that each sensor is connected and reporting reasonable values before the car goes on track.

Calibration means checking a sensor against a known condition and correcting its setup where necessary. An IMU can be checked while the car is stationary, wheel speed can be compared with a known rotation count, and cameras need calibration so software understands their lens and position. GNSS antennas should be installed in a consistent location, while LiDAR position and angle should be measured relative to the car. Calibration details should be saved with the test data. If a sensor is moved or replaced, the team should repeat the relevant checks.

5. Cost and Sourcing in India

Prices vary by seller, availability, specification and shipping, so the figures in this report are planning estimates rather than guaranteed quotations. Online Indian listings provide examples of low-cost IMU boards from about ₹140 to ₹1,500 and basic GNSS modules from about ₹400 to ₹1,300. Small LiDAR distance modules can be around ₹1,800–₹4,000, while scanning LiDAR can range from roughly ₹8,000 to above ₹50,000. RTK GNSS, stereo cameras and race-grade wheel speed sensors can be substantially more expensive. A quoted price may not include all cables, brackets, connectors, data-acquisition hardware, software or import charges.

For basic electronics, the team can check Indian online sellers such as Roboarc, Zbotic, Indian Hobby Center, Robu.in and other established component stores. Mouser India and DigiKey India are useful for branded electronic components, while specialist robotics and drone suppliers may stock GNSS and LiDAR equipment. Camera manufacturers such as Stereolabs sell stereo cameras directly, though delivery, taxes and warranty conditions should be checked. These examples are starting points for research, not endorsements; the team should compare part numbers and seller terms before ordering. Prefer a seller that provides a proper invoice, a clear return policy and technical documentation.

Before placing an order, ask the seller for a written quotation that includes taxes, shipping, delivery time and warranty. Check whether the product is in stock or only available on backorder. Confirm the exact model and interface, because similar-looking sensor boards may use different chips or communication protocols. For imported products, ask about customs charges and whether the seller will handle them. Keep a record of the product page, quotation and datasheet so the team can repeat the purchase later. Buying one sample first is often safer than ordering several units before testing compatibility.

A. *Cost Planning and Buying Risks*

A useful purchase plan separates essential items from optional upgrades. The essential list may include an IMU, wheel speed sensors, connectors and a logger. GNSS can be added when route and lap-position data are needed. Cameras and LiDAR are more relevant when the car must recognise track features without relying on a driver. The team should also estimate the cost of spares because a damaged connector or sensor can delay testing. It is sensible to keep one spare of low-cost critical sensors when the budget allows.

The cheapest online listing is not always the lowest-cost choice overall. A board may arrive without a cable, use an unfamiliar connector or lack clear documentation. Some products are intended for classroom experiments rather than repeated exposure to vibration, heat and dust. Before ordering, read the datasheet and confirm the supply voltage, output format, measurement range and operating temperature. For high-value sensors, request a quotation and check the return and warranty terms. The listed prices in this report are examples seen during research and should be verified again at the time of purchase.

B. Example Sourcing and Indicative Prices

Indian hobby-electronics stores list basic motion and satellite modules at relatively low prices. For example, Roboarc has listed MPU-6050 and BNO055-type boards, while Indian Hobby Center lists accelerometer and encoder-related modules. Zbotic and other robotics suppliers list small LiDAR distance modules as well as scanning units. Online examples include a BNO055 board around ₹1,500, a basic NEO-6M GNSS module around ₹400, a NEO-M8N module around ₹1,300, and small TF-Luna or TFMini-style LiDAR modules from roughly ₹1,800 upward. Prices and availability can change, so these figures should be treated as a starting budget only.

For more demanding use, the cost rises quickly. A stereo camera such as the ZED 2i includes two cameras and depth-processing features, so it costs far more than a simple USB camera. RTK GNSS requires a compatible receiver, antenna and correction source, and scanning LiDAR costs more than a single-point distance module. These systems may also need a more capable computer and additional mounting hardware. The team should compare total system cost, not just the price of the sensor itself.

6. Suggested Purchase Plan

The team can build its sensor package in stages. First, install an IMU, wheel speed sensors and selected suspension or pedal sensors to support vehicle testing and data collection. These give useful information even without a driverless system. Next, add GNSS to record the route and compare lap-to-lap behaviour. If the project includes autonomous driving, test a mono camera first for cone detection, then compare it

with a stereo camera and LiDAR based on the accuracy and processing power required. The team should choose the smallest set of sensors that answers its key questions.

The sensor prices are only part of the total cost. The team should also budget for cables, brackets, connectors, protective housings, power supplies, a suitable computer and time needed for installation and testing. Each sensor must use a compatible voltage and communication method, and its data must be recorded with a consistent timestamp. A sensor that is accurate but produces missing or delayed data may not be useful for vehicle analysis. During testing, check for loose mounts, noisy signals, overheating and readings that jump unexpectedly. Write down calibration steps so results can be compared between practice days.

A. Testing Plan Before Competition

Testing should begin with the car stationary, then continue at low speed in a safe, controlled area before normal practice runs. First check that every sensor gives a sensible reading and that cables remain secure. Next compare sensor readings during simple actions such as turning slowly, accelerating and braking. Once the system works consistently, collect data over repeated laps and compare similar sections of the track. Changes to the sensor position, software or calibration should be noted so that one run can be compared fairly with another.

The team should also define what success means for each sensor. For wheel speed, this could be stable readings at different speeds without missed pulses. For GNSS, it could be continuous position data with a clear view of the sky. For a camera or LiDAR, it could be consistent detection of track cones at the distances required by the vehicle. These checks help identify whether a problem comes from the sensor, mounting, wiring, software or environmental conditions. Clear test records will make future maintenance and sensor upgrades easier.

7. Conclusion

The most useful sensor choices depend on the car's purpose. An IMU measures movement, GNSS provides position, wheel sensors measure rotation, cameras help identify cones, and LiDAR measures distance to nearby objects. Suspension, steering, brake and temperature sensors add detail that can help the team improve the vehicle setup. No single sensor gives a complete picture: combining readings can make the results more useful, but only if the sensors are mounted properly, calibrated and checked regularly.

Before finalising purchases, test the IMU under vibration, check GNSS reception at the practice site and try the camera and LiDAR with real track cones. Record the sensor's limitations as well as its strengths. Compare supplier prices again before ordering because online listings and stock can change. A simple, well-tested sensor system is more useful than a complex system that the team cannot calibrate or maintain. The final selection should balance cost, reliability, ease of installation and the type of data needed for Formula Student testing.

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